PH Weather Advisory — Team Guide

Nov 7, 2026 · @khair

Project at a glance

We are building a weather app for the Philippines that gives simple daily advice in Taglish or English. Real forecast numbers decide what the app says, and AI explains them in friendly words.

The first version has 4 features:

- **7-day forecast:** pick a city and see the week ahead
- **Payo for today:** 2 to 3 sentences of advice for the day
- **Alis-Uwi:** set your leave and return time, see if it will rain
- **Heat check:** how hot it will really feel

One rule above all: the numbers come from code, never from the AI. The AI only explains.

Meet the team

Seven people, seven parts. Each person owns one part of the app and has a backup who can help.

Name	Role	In one line	Works closest with	Backup
Khair	Project Manager and Tech Lead	Keeps the project moving and reviews all work	Everyone	—
Joan	UI/UX Designer	Designs how the app looks and feels	Miguel	Miguel
Miguel	Frontend Developer	Builds the screens people tap	Joan, Fredelina	Joan
MJ	Data Engineer	Brings in the weather data and stores it	Fredelina, Angela	Fredelina
Fredelina	Backend Developer	Turns data into facts and sends them to the app	MJ, JC, Miguel	JC

Name	Role	In one line	Works closest with	Backup
JC	AI Integrator	Turns facts into friendly advice	Fredelina, Angela	Fredelina
Angela	QA and Docs	Tests everything and keeps the docs	Everyone	—

Who does what

Each role below lists what you own and who you hand your work to.

Khair — Project Manager and Tech Lead

Leads the team, decides how the app is built, and reviews all work before it goes in.

- Set up the repo, folders and starter code so everyone can start right away
- Write the cards and keep the board updated
- Review every pull request, ideally within a day
- Keep the API keys and set the spending limit
- Run the weekly planning and the weekly demo
- Check in when a card sits in In Progress for more than 2 days

Joan — UI/UX Designer

Designs how the app looks and how people move around in it.

- Sketch the screens first: City Picker, 7-day forecast, Payo for today, Alis-Uwi, heat card
- Make the final screens in Figma
- Write a short style guide in DESIGN.md: colors, fonts, spacing, buttons
- Check the built screens against the design and point out what is off
- Help with styling once the designs are done

Hands off to Miguel: finished screens with notes on spacing and colors.

Miguel — Frontend Developer

Builds the screens people see and tap, and connects them to the backend.

- Build the city picker and the 7-day forecast
- Build the Alis-Uwi screen for leave and return times
- Build the heat card and the Payo for today card
- Show a simple message when something fails or the signal is weak

Gets: designs from Joan, API endpoints from Fredelina.

MJ — Data Engineer

Brings the weather data into the app and keeps it clean and saved.

- Get the hourly forecast from Open-Meteo
- Keep the list of cities and their locations for the city picker
- Save forecasts in the database so the same data is not fetched again and again
- Refresh the data on a schedule, for example every hour
- Check for missing or strange values before passing data on

Hands off to Fredelina: clean, saved forecasts ready to use.

Fredelina — Backend Developer

Turns the weather data into clear facts and serves them to the app.

- Write the rain check and the heat level as plain code
- Build the API endpoints the frontend calls
- Put the facts together (rain by hour, heat level, leave and return times) and send them to JC's advice function

Gets: data from MJ. **Hands off:** endpoints to Miguel, facts to JC.

JC — AI Integrator

Turns the facts into short, friendly advice in Taglish or English. Works inside the backend code.

- Set up OpenRouter and try 2 or 3 cheap models to find the best Taglish
- Write the prompts using basic LangChain
- Build the advice function: facts and language in, 2 to 3 sentences out
- Add a backup message for when the AI fails or takes too long
- Keep a list of sample forecasts and the advice each should give

Remember: agree on the function format with Fredelina on day one. The severe weather banner is fixed text, not AI. The API key stays on the server, never in the repo or frontend.

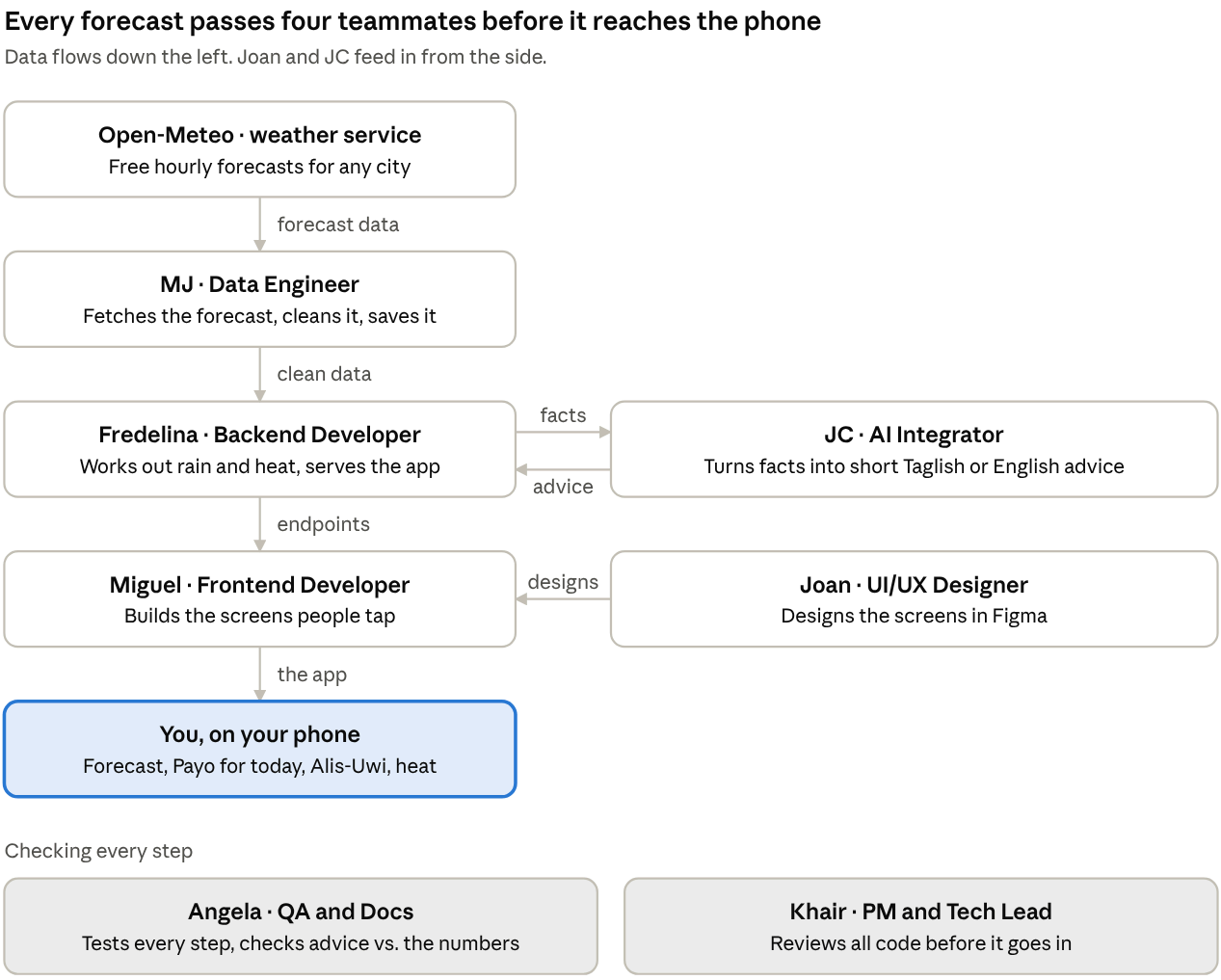
Angela — QA and Docs

Tests the app and keeps the documents. Makes sure what the app says matches the real forecast.

- Compare every AI advice against the raw numbers: if it says rain, the rain chance should really be high
- Test each feature once built and report bugs on the board
- Retest bugs after they are fixed
- Write and update the README
- Keep notes for the weekly demo

Sends: each bug to the person who owns that part. This role touches every part, so it is a great way to learn the whole app.

How the app works

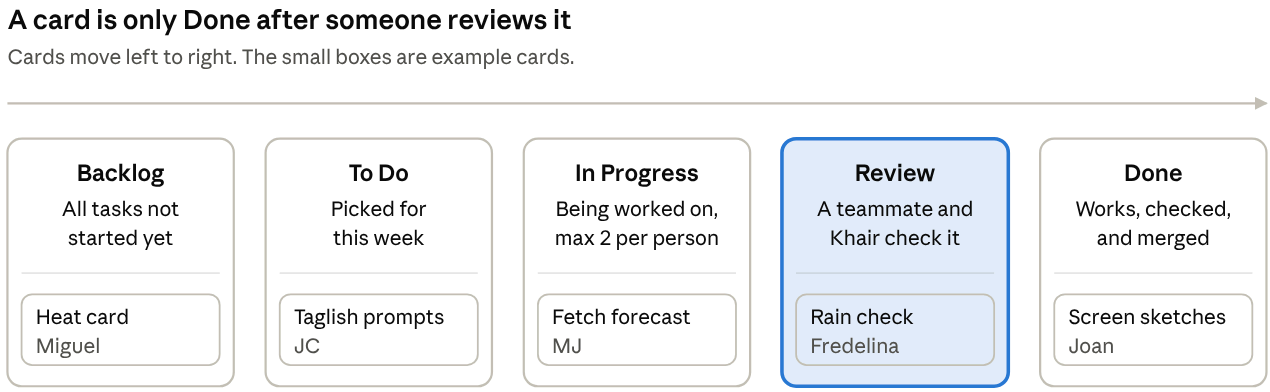


data flow · who hands what to whom

MJ brings the data in, Fredelina turns it into facts, JC adds the words, and Miguel shows it on screen. Joan designs the screens, and Angela and Khair check every step.

How we work: Agile Kanban

Every task is a card that moves across one board. We build in small pieces, show our work every week, and improve as we go.

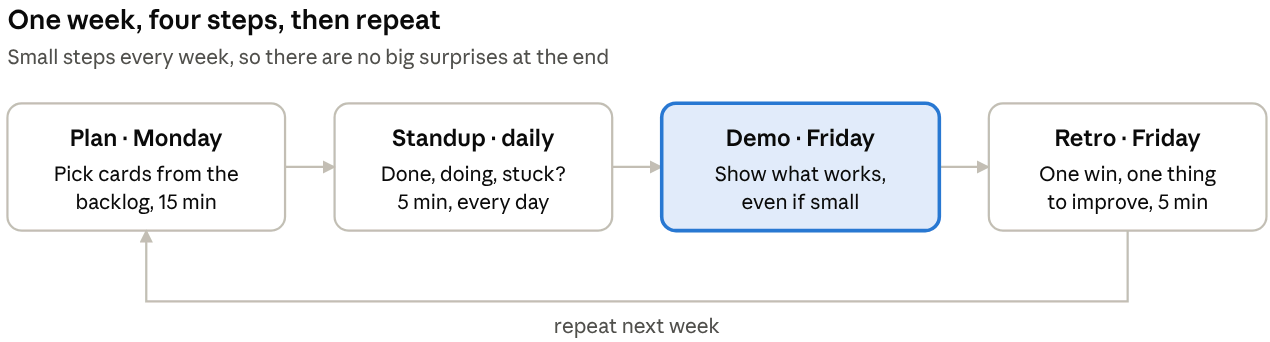


our Kanban board · 5 columns

Board rules

- Keep cards small: 1 to 2 days of work
- Every card has a "Done when…" line so you know when to stop
- Max 2 cards In Progress per person
- One Git branch per card. Open a pull request to move it to Review.

The weekly loop



the weekly loop · 4 steps

The Friday demo is the most important step. Show what works, even if it is small.

The life of one card

Each card goes through a mini version of how all software is built:

1. **Understand:** read the card and its "Done when" line
2. **Plan:** sketch or think through how you'll do it

3. **Build:** write the code on your own branch
4. **Test:** check that it works
5. **Review:** open a pull request so a teammate and Khair can check it

Tools we'll use

All tools below are free or have a free plan, and are friendly for beginners. **Tools may change as the project grows.** If a tool is not working for you, bring it up at weekly planning.

Area	Tool	What it's for	Main user
Everyone	VS Code	Writing code	All
Everyone	Git and GitHub	Saving and sharing code, pull requests	All
Everyone	GitHub Projects	Our Kanban board	All
Design	Google Stitch	Quick screen ideas to start from	Joan
Design	Figma	Final screen designs	Joan
Design	DESIGN.md	Our style guide as a simple text file	Joan, Miguel
Frontend	React with Vite	Building the screens	Miguel
Backend	Python with FastAPI	Building the API, with a built-in page to try endpoints	Fredelina, JC
Data	Open-Meteo, NASA	Free weather forecast data, no key needed	MJ
Data	PostgreSQL and SQL	Saving forecasts and cities	MJ, Fredelina
Data	MongoDB	Option for saving flexible data like advice samples	MJ, JC
Data	pgAdmin	Looking at the database tables	MJ
AI	OpenRouter	One key to try many AI models	JC
AI	LangChain (basic)	Writing and managing prompts	JC

Area	Tool	What it's for	Main user
Testing	pytest	Automatic checks for backend code	Angela, Fredelina
Testing	Postman, Swagger	Trying API endpoints by hand	Angela
Docs	Markdown	Writing the README and notes	Angela

What you'll learn

Every lesson connects to a real part of this app. You'll use some topics every day and see the rest when roles rotate.

Lesson	Where you use it in this project
Git and GitHub	Everyone, every day: branches, commits, pull requests
Python	Backend, data and AI code
React	The app's screens
Stitch and design systems (DESIGN.md)	Planning screens and keeping the look the same everywhere
SQL, PostgreSQL and MongoDB	Saving and reading forecasts and cities
Prompt engineering	Getting the AI to write short, correct Taglish advice
Agent modelling	Planning how the AI part takes facts in and gives advice out
Testing	Checking every feature before it counts as done

Team rules

1. **Stuck for 30 minutes?** Ask a teammate first. Still stuck, ask Khair.
2. **Everyone has a backup:** Joan and Miguel cover the UI. Fredelina, JC and MJ cover the server and data.
3. **Roles rotate at the halfway point** so everyone tries something new.

4. **Keep the first version small:** forecast, Payo for today, Alis-Uwi and heat check only.
5. **Numbers come from code, never from the AI.**
6. **No API keys in the repo or the frontend.** Ever.
7. **Not knowing is normal.** Ask early, show your work often.

Words to know

Word	Simple meaning
Repo	The project folder on GitHub where all our code lives
Branch	Your own copy of the code to work on one card safely
Pull request (PR)	Asking the team to check and add your work
API	How two programs talk to each other
Endpoint	One address in the API, like a door that gives back one kind of data
Card, Tickets	One small task on the board
Backlog	The list of all tasks we have not started
Standup	A quick daily check-in